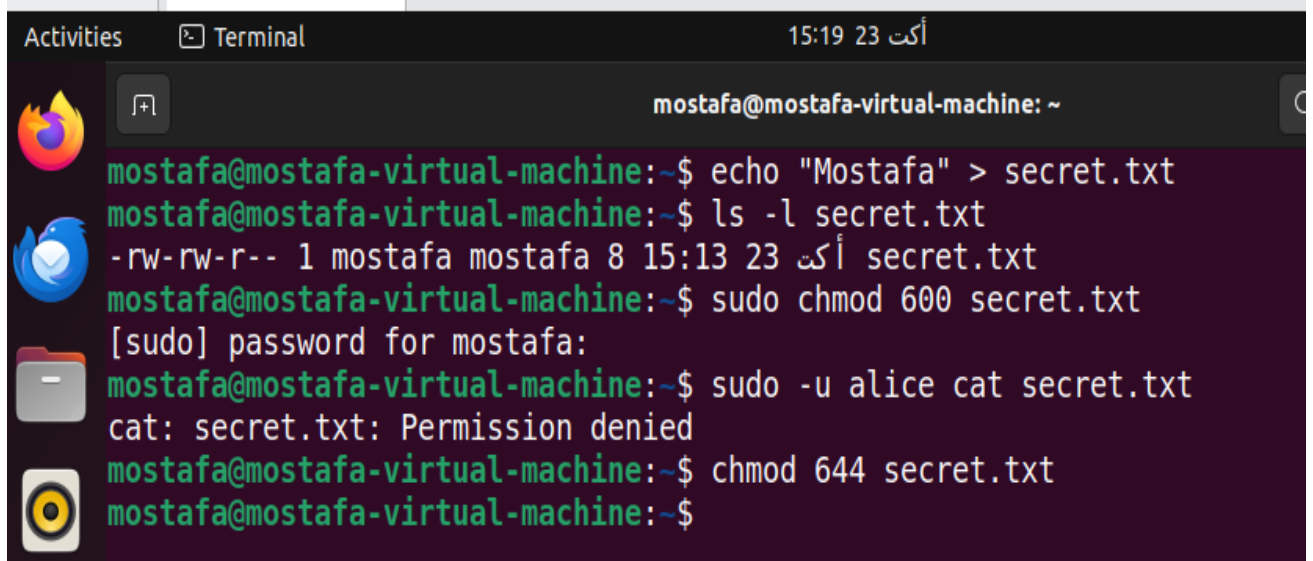


Name : Mostafa Ehab Diab Nassar

Exercise 1: Permissions

1. Create a file called `secret.txt` with some content
2. Check its permissions with `ls -l`
3. Change permissions to 600 (owner read/write only)
4. Try to read it as another user
5. Change permissions back to 644

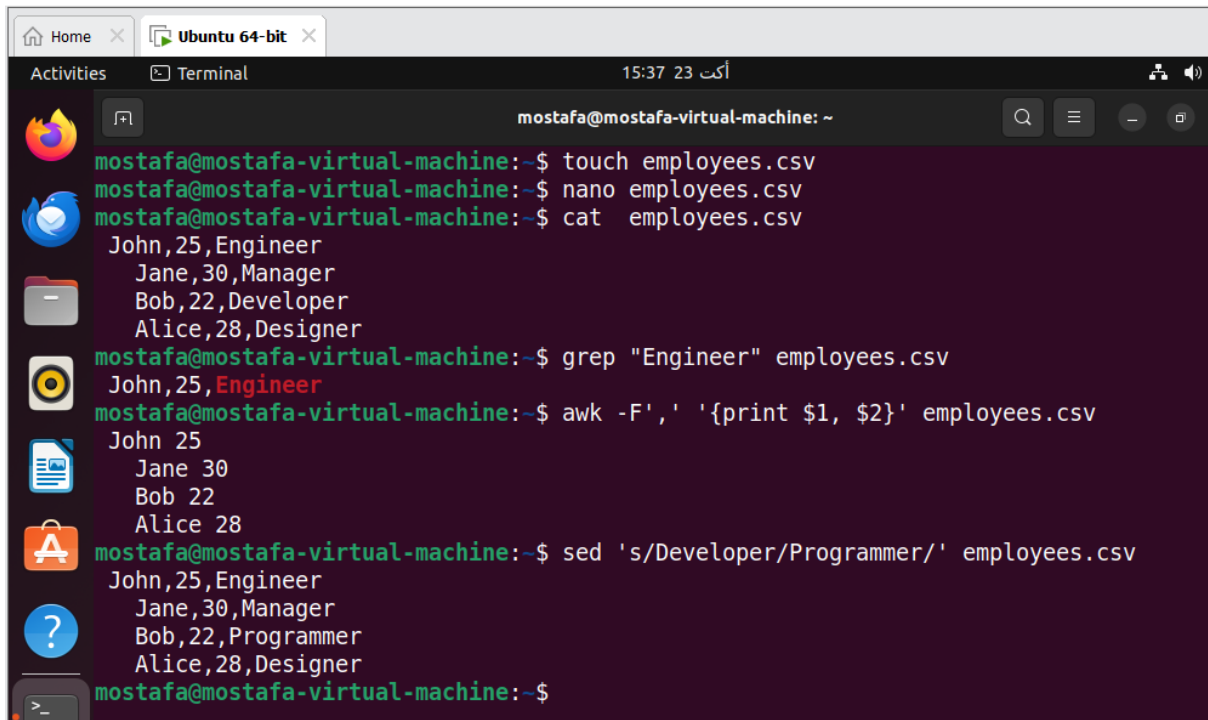
A terminal window titled 'Terminal' with a clock showing 15:19 and the date 23 أكت. The terminal shows the user 'mostafa' at 'mostafa-virtual-machine'. The commands and output are: 1. 'echo "Mostafa" > secret.txt' creates the file. 2. 'ls -l secret.txt' shows permissions '-rw-rw-r--'. 3. 'sudo chmod 600 secret.txt' changes permissions. 4. 'sudo -u alice cat secret.txt' fails with 'Permission denied'. 5. 'chmod 644 secret.txt' changes permissions back. The prompt returns to the user.

```
mostafa@mostafa-virtual-machine:~$ echo "Mostafa" > secret.txt
mostafa@mostafa-virtual-machine:~$ ls -l secret.txt
-rw-rw-r-- 1 mostafa mostafa 8 15:13 23 أكت secret.txt
mostafa@mostafa-virtual-machine:~$ sudo chmod 600 secret.txt
[sudo] password for mostafa:
mostafa@mostafa-virtual-machine:~$ sudo -u alice cat secret.txt
cat: secret.txt: Permission denied
mostafa@mostafa-virtual-machine:~$ chmod 644 secret.txt
mostafa@mostafa-virtual-machine:~$
```

Exercise 2: Text Processing

1. Create a file with the following content:
John,25,Engineer
Jane,30,Manager
Bob,22,Developer
Alice,28,Designer
2. Use `grep` to find lines containing "Engineer"
3. Use `awk` to print only names and ages

4. Use `sed` to change "Developer" to "Programmer"



```
mostafa@mostafa-virtual-machine: ~  
mostafa@mostafa-virtual-machine:~$ touch employees.csv  
mostafa@mostafa-virtual-machine:~$ nano employees.csv  
mostafa@mostafa-virtual-machine:~$ cat employees.csv  
John,25,Engineer  
Jane,30,Manager  
Bob,22,Developer  
Alice,28,Designer  
mostafa@mostafa-virtual-machine:~$ grep "Engineer" employees.csv  
John,25,Engineer  
mostafa@mostafa-virtual-machine:~$ awk -F',' '{print $1, $2}' employees.csv  
John 25  
Jane 30  
Bob 22  
Alice 28  
mostafa@mostafa-virtual-machine:~$ sed 's/Developer/Programmer/' employees.csv  
John,25,Engineer  
Jane,30,Manager  
Bob,22,Programmer  
Alice,28,Designer  
mostafa@mostafa-virtual-machine:~$
```

```bash

# Print names and ages only

```
awk -F',' '{print $1, $2}' employees.csv
```

# Find people older than 28

```
awk -F',' '$2 > 28' employees.csv
```

# Calculate average age

```
awk -F',' '{sum += $2} END {print "Average age:", sum/NR}' employees.csv
```

# Count people by position

```
awk -F',' '{count[$3]++} END {for (i in count) print i, count[i]}' employees.csv
```

# Add line numbers and field count

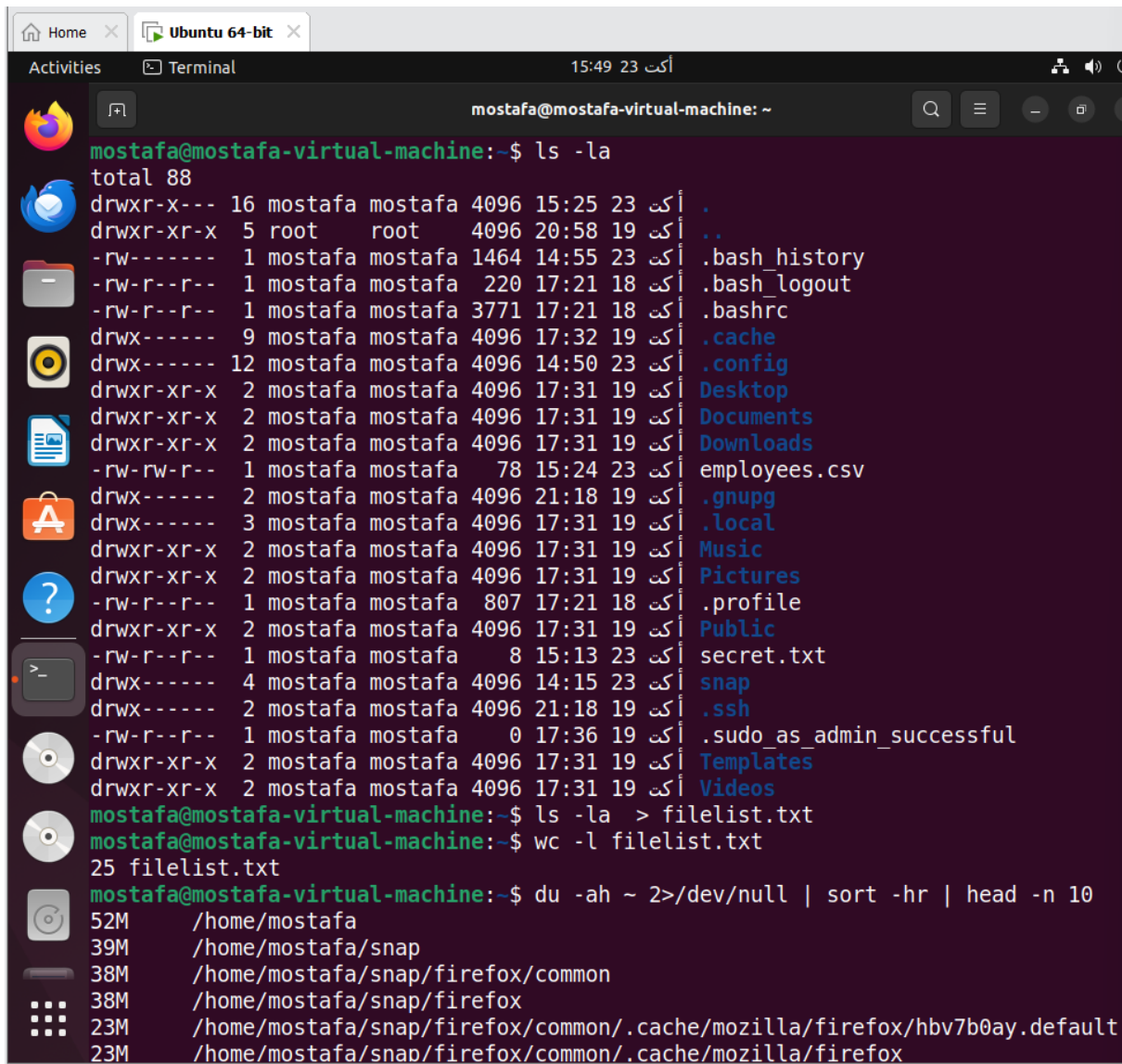
```
awk -F',' '{print NR, NF, $0}' employees.csv
```

```
mostafa@mostafa-virtual-machine:~$ awk -F',' '{print $1, $2}' employees.csv
John 25
Jane 30
Bob 22
Alice 28
mostafa@mostafa-virtual-machine:~$ awk -F',' '$2 > 28' employees.csv
Jane,30,Manager
mostafa@mostafa-virtual-machine:~$ awk -F',' '{sum += $2} END {print "Average age:",
sum/NR}' employees.csv
Average age: 26.25
mostafa@mostafa-virtual-machine:~$ awk -F',' '{count[$3]++} END {for (i in count) pri
nt i, count[i]}' employees.csv
Designer 1
Manager 1
Developer 1
Engineer 1
mostafa@mostafa-virtual-machine:~$ awk -F',' '{print NR, NF, $0}' employees.csv
1 3 John,25,Engineer
2 3 Jane,30,Manager
3 3 Bob,22,Developer
4 3 Alice,28,Designer
mostafa@mostafa-virtual-machine:~$
```

...

### ### Exercise 3: Pipes and Redirection

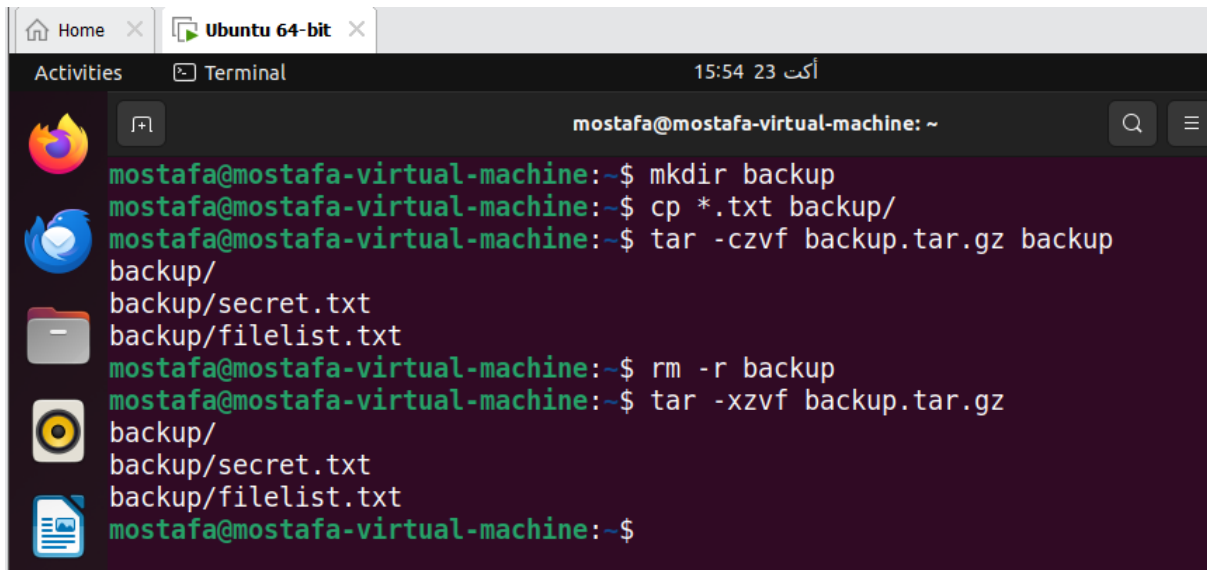
1. List all files in your home directory
2. Save the list to a file called `filelist.txt`
3. Count how many files you have
4. Find the largest file in your home directory



```
mostafa@mostafa-virtual-machine: ~
mostafa@mostafa-virtual-machine:~$ ls -la
total 88
drwxr-x--- 16 mostafa mostafa 4096 15:25 23 أك .
drwxr-xr-x 5 root root 4096 20:58 19 أك ..
-rw----- 1 mostafa mostafa 1464 14:55 23 أك .bash_history
-rw-r--r-- 1 mostafa mostafa 220 17:21 18 أك .bash_logout
-rw-r--r-- 1 mostafa mostafa 3771 17:21 18 أك .bashrc
drwx----- 9 mostafa mostafa 4096 17:32 19 أك .cache
drwx----- 12 mostafa mostafa 4096 14:50 23 أك .config
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Desktop
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Documents
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Downloads
-rw-rw-r-- 1 mostafa mostafa 78 15:24 23 أك employees.csv
drwx----- 2 mostafa mostafa 4096 21:18 19 أك .gnupg
drwx----- 3 mostafa mostafa 4096 17:31 19 أك .local
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Music
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Pictures
-rw-r--r-- 1 mostafa mostafa 807 17:21 18 أك .profile
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Public
-rw-r--r-- 1 mostafa mostafa 8 15:13 23 أك secret.txt
drwx----- 4 mostafa mostafa 4096 14:15 23 أك snap
drwx----- 2 mostafa mostafa 4096 21:18 19 أك .ssh
-rw-r--r-- 1 mostafa mostafa 0 17:36 19 أك .sudo_as_admin_successful
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Templates
drwxr-xr-x 2 mostafa mostafa 4096 17:31 19 أك Videos
mostafa@mostafa-virtual-machine:~$ ls -la > filelist.txt
mostafa@mostafa-virtual-machine:~$ wc -l filelist.txt
25 filelist.txt
mostafa@mostafa-virtual-machine:~$ du -ah ~ 2>/dev/null | sort -hr | head -n 10
52M /home/mostafa
39M /home/mostafa/snap
38M /home/mostafa/snap/firefox/common
38M /home/mostafa/snap/firefox
23M /home/mostafa/snap/firefox/common/.cache/mozilla/firefox/hbv7b0ay.default
23M /home/mostafa/snap/firefox/common/.cache/mozilla/firefox
```

### ### Exercise 4: File Operations

1. Create a directory called `backup`
2. Copy some files to the backup directory
3. Create a tar archive of the backup directory
4. Remove the original backup directory
5. Extract the archive to verify it works

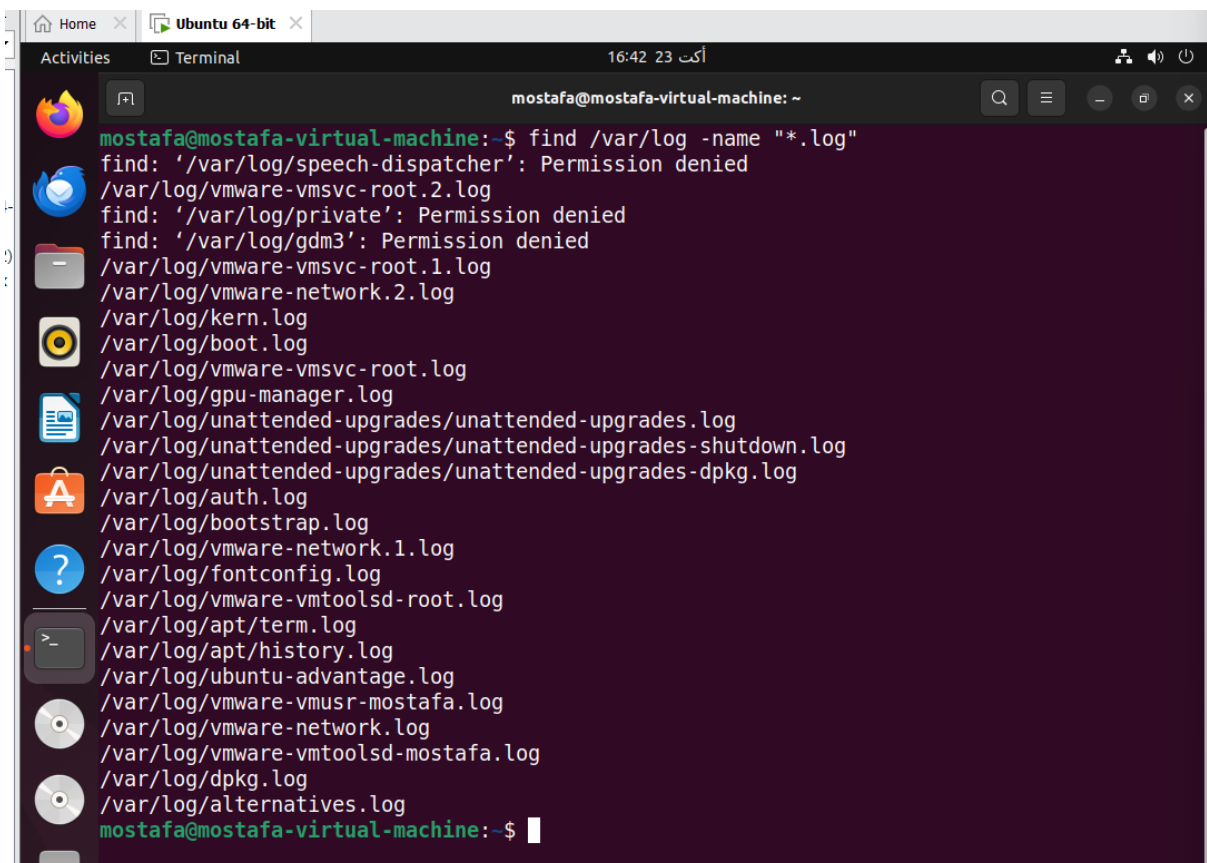
A terminal window titled 'Ubuntu 64-bit' with a search bar and a menu icon. The prompt is 'mostafa@mostafa-virtual-machine: ~'. The user enters a series of commands to create a backup directory, copy text files, create a tar archive, remove the directory, and then extract the archive. The output shows the files being processed.

```
mostafa@mostafa-virtual-machine:~$ mkdir backup
mostafa@mostafa-virtual-machine:~$ cp *.txt backup/
mostafa@mostafa-virtual-machine:~$ tar -czvf backup.tar.gz backup
backup/
backup/secret.txt
backup/filelist.txt
mostafa@mostafa-virtual-machine:~$ rm -r backup
mostafa@mostafa-virtual-machine:~$ tar -xzvf backup.tar.gz
backup/
backup/secret.txt
backup/filelist.txt
mostafa@mostafa-virtual-machine:~$
```

## ## 9. Homework

1. Create a script that:

- Finds all `.log` files in `/var/log`

A terminal window titled 'Ubuntu 64-bit' with a search bar and a menu icon. The prompt is 'mostafa@mostafa-virtual-machine: ~'. The user enters the command 'find /var/log -name "\*.log"'. The output lists various log files in /var/log, including some with permission denied messages.

```
mostafa@mostafa-virtual-machine:~$ find /var/log -name "*.log"
find: '/var/log/speech-dispatcher': Permission denied
/var/log/vmware-vmtoolsd-root.2.log
find: '/var/log/private': Permission denied
find: '/var/log/gdm3': Permission denied
/var/log/vmware-vmtoolsd-root.1.log
/var/log/vmware-network.2.log
/var/log/kern.log
/var/log/boot.log
/var/log/vmware-vmtoolsd-root.log
/var/log/gpu-manager.log
/var/log/unattended-upgrades/unattended-upgrades.log
/var/log/unattended-upgrades/unattended-upgrades-shutdown.log
/var/log/unattended-upgrades/unattended-upgrades-dpkg.log
/var/log/auth.log
/var/log/bootstrap.log
/var/log/vmware-network.1.log
/var/log/fontconfig.log
/var/log/vmware-vmtoolsd-root.log
/var/log/apt/term.log
/var/log/apt/history.log
/var/log/ubuntu-advantage.log
/var/log/vmware-vmtoolsd-mostafa.log
/var/log/vmware-network.log
/var/log/vmware-vmtoolsd-mostafa.log
/var/log/dpkg.log
/var/log/alternatives.log
mostafa@mostafa-virtual-machine:~$
```

- Searches for "error" in each file

```
/var/log/alternatives.log
mostafa@mostafa-virtual-machine:~$ grep -r "error" /var/log
grep: /var/log/speech-dispatcher: Permission denied
grep: /var/log/btmp: Permission denied
grep: /var/log/vmware-vmtoolsd-root.2.log: Permission denied
/var/log/syslog:Oct 19 16:42:17 mostafa-virtual-machine systemd[1]: Condition check resulted in Process error reports when automatic reporting is enabled (file watch) being skipped.
/var/log/syslog:Oct 19 16:42:17 mostafa-virtual-machine systemd[1]: Condition check resulted in Process error reports when automatic reporting is enabled (timer based) being skipped.
/var/log/syslog:Oct 19 16:42:17 mostafa-virtual-machine alsactl[804]: alsa-lib main.c:1412:(snd_use_case_mgr_open) error: failed to import hw:0 use case configuration -2
/var/log/syslog:Oct 19 16:42:20 mostafa-virtual-machine gnome-session-binary[910]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
/var/log/syslog:Oct 19 16:42:20 mostafa-virtual-machine gnome-session-binary[910]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
/var/log/syslog:Oct 19 16:42:20 mostafa-virtual-machine gnome-session-binary[910]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
/var/log/syslog:Oct 19 16:42:20 mostafa-virtual-machine gnome-session-binary[910]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
/var/log/syslog:Oct 19 17:31:10 mostafa-virtual-machine systemd[1]: Condition check resulted in Process error reports when automatic reporting is enabled (file watch) being skipped.
/var/log/syslog:Oct 19 17:31:10 mostafa-virtual-machine systemd[1]: Condition check resulted in Process error reports when automatic reporting is enabled (timer based) being skipped.
/var/log/syslog:Oct 19 17:31:11 mostafa-virtual-machine alsactl[799]: alsa-lib main.c:1412:(snd_use_case_mgr_open) error: failed to import hw:0 use case configuration -2
/var/log/syslog:Oct 19 17:31:12 mostafa-virtual-machine gnome-session-binary[976]: GLib-GIO-CRITICAL: g_bus_get_sync: assertion 'error == NULL || *error == NULL' failed
```

- Saves the results to `error\_report.txt`

## 2. Practice vim by creating and editing a file

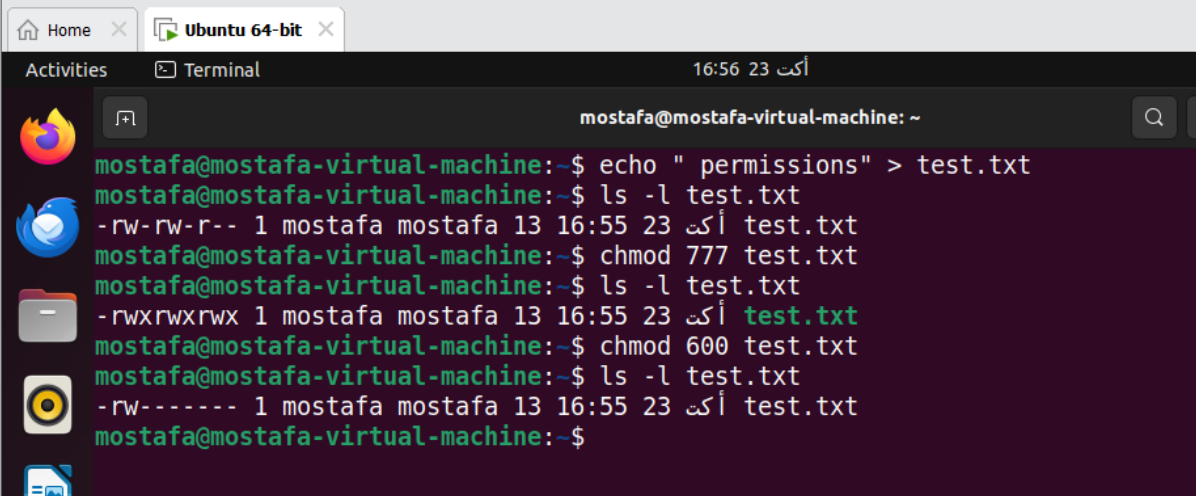
```
mostafa@mostafa-virtual-machine:~$ vim practice.txt
mostafa@mostafa-virtual-machine:~$ cat practice.txt
Hello

mostafa@mostafa-virtual-machine:~$
```

## 3. Create a backup of your important files using tar

```
mostafa@mostafa-virtual-machine:~$ mkdir myfiles
mostafa@mostafa-virtual-machine:~$ echo "Backup" > myfiles/info.txt
mostafa@mostafa-virtual-machine:~$ cp /etc/hosts myfiles/
mostafa@mostafa-virtual-machine:~$ tar -czvf mybackup.tar.gz myfiles
myfiles/
myfiles/info.txt
myfiles/hosts
mostafa@mostafa-virtual-machine:~$ ls -lh mybackup.tar.gz
-rw-rw-r-- 1 mostafa mostafa 338 16:53 23 أكتوبر mybackup.tar.gz
mostafa@mostafa-virtual-machine:~$
```

#### 4. Experiment with different permission combinations



```
mostafa@mostafa-virtual-machine: ~
mostafa@mostafa-virtual-machine:~$ echo " permissions" > test.txt
mostafa@mostafa-virtual-machine:~$ ls -l test.txt
-rw-rw-r-- 1 mostafa mostafa 13 16:55 23 أكت test.txt
mostafa@mostafa-virtual-machine:~$ chmod 777 test.txt
mostafa@mostafa-virtual-machine:~$ ls -l test.txt
-rwxrwxrwx 1 mostafa mostafa 13 16:55 23 أكت test.txt
mostafa@mostafa-virtual-machine:~$ chmod 600 test.txt
mostafa@mostafa-virtual-machine:~$ ls -l test.txt
-rw----- 1 mostafa mostafa 13 16:55 23 أكت test.txt
mostafa@mostafa-virtual-machine:~$
```